EXPERIMENT 4:

MEASUREMENT AND ANALYSIS OF DISPERSION COMPENSATION IN OPTICAL FIBER USING OPTISYSTEM

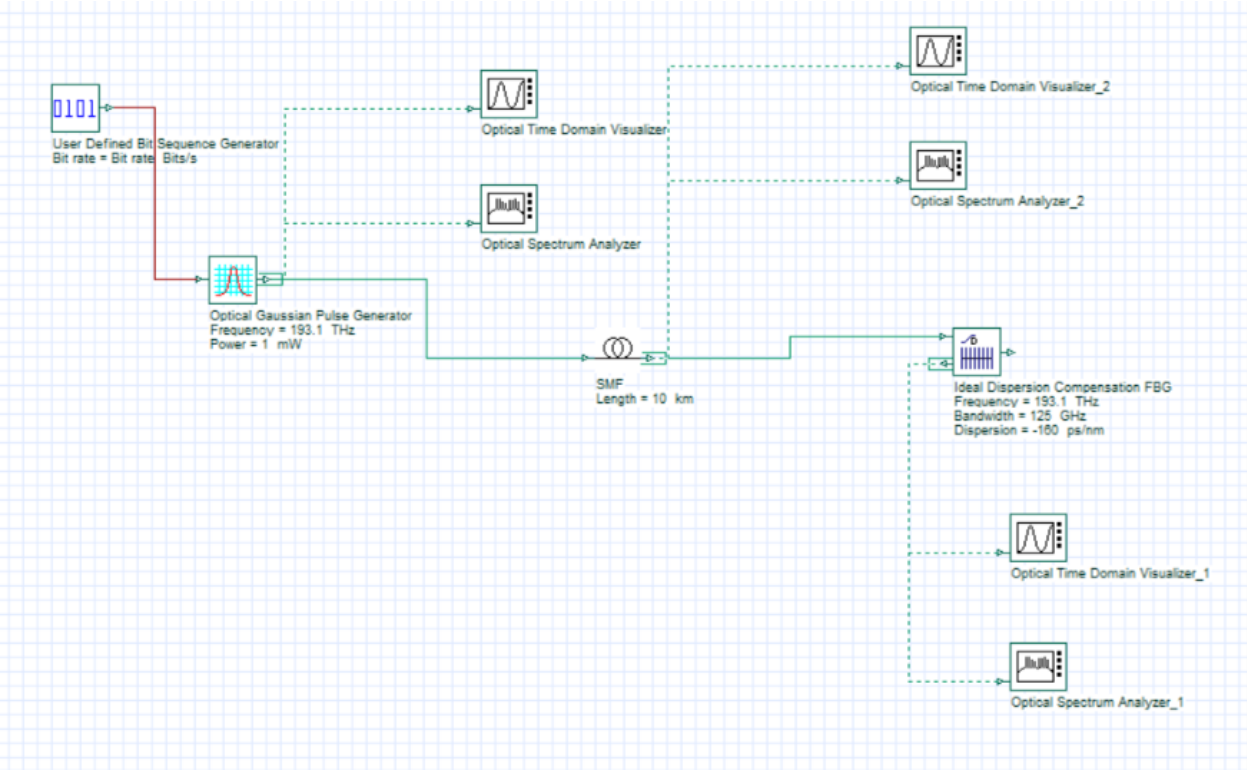
Objective :

design and simulate an optical fiber communication system using Opti System, observe pulse broadening due to fiber dispersion, and verify pulse recovery using a dispersion compensation component.

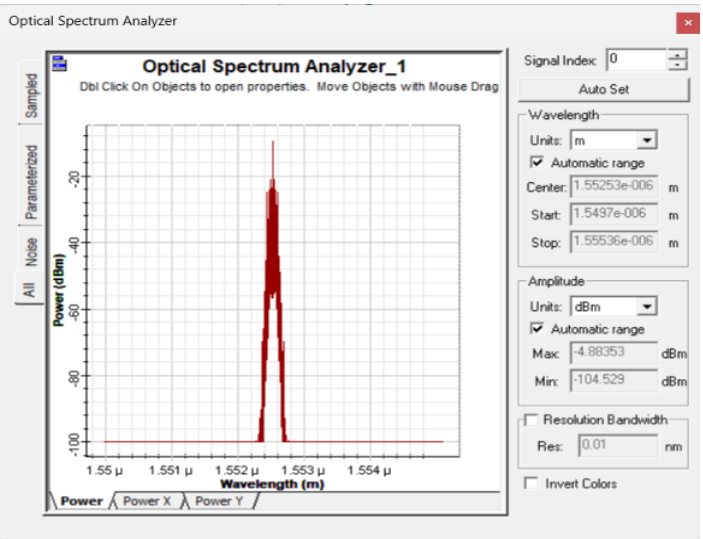
SOFTWARE REQUIRED:

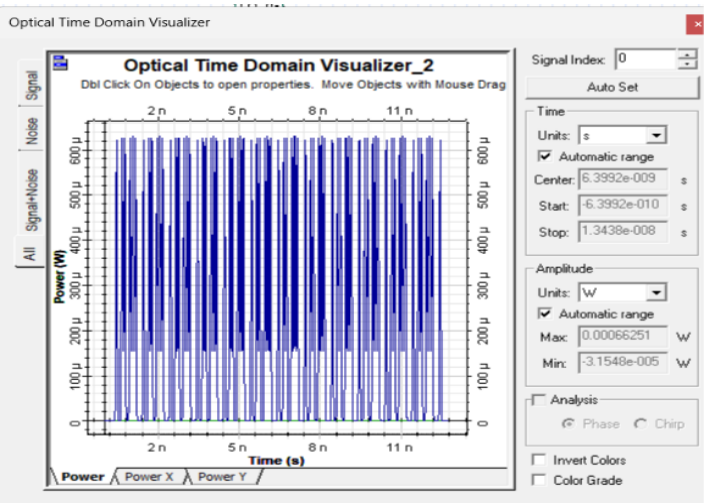
OptiSystem.

circuit DIAGRAM:



OUTPUTS:





REFLECTION:

- I learned how dispersion affects optical signals during transmission.
- I observed pulse broadening in the single-mode fiber.
- I understood how a dispersion compensation FBG reduces pulse broadening.
- I learned to analyze optical signals using OptiSystem visualizers.

RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Design the OptiSystem layout	Project layout	Builds the complete layout with correct components and connections independently	Minor connection errors corrected	Needs guidance for components or connections	Unable to build the layout	20
Configure pulse, SMF and compensator parameters	Parameter configuration	Sets all parameters (40 Gb/s, 12.5 ps, 193.1 THz, 10 km, −160 ps/nm, 125 GHz) correctly	Minor parameter errors	Needs guidance for several parameters	Unable to configure parameters	20
Observe pulse broadening in SMF	Pulse observation	Captures and interprets initial and broadened pulses correctly	Minor interpretation errors	Limited observation / needs help	Unable to obtain or interpret pulses	20
Verify compensation using ideal dispersion component	Compensation verification	Verifies recovery of the pulse and explains residual effects of settings	Mostly correct verification	Partial verification with assistance	Unable to verify compensation	20
Document findings	Report and conclusion	Complete report with justified conclusion	Minor omissions	Incomplete record / weak conclusion	Report / conclusion inadequate	20

CONCLUSION :

1. The optical fiber communication system was designed and simulated successfully using OptiSystem.
2. Pulse broadening due to fiber dispersion was observed in the time-domain visualizer.
3. The dispersion compensation FBG was configured with a dispersion value of −160 ps/nm to compensate for the accumulated fiber dispersion.
4. The optical spectrum and time-domain outputs were observed before and after compensation to study the effect of dispersion compensation.